<u>Page 1 – Cover / Intro Page</u>

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Pole Switching (Dahlander) Motor Control – Experiment Report

This experiment is done to understand how a Dahlander-type three-phase induction motor works when its poles are changed to get two different speeds. A Dahlander motor has a special winding that can be connected in two ways. When the number of poles is high, the motor runs at low speed, and when the poles are reduced, the motor runs at high speed. This helps us study how speed, current, power, and torque change when the motor operates in both modes.

For performing the experiment, we used a Dahlander motor, a 3-phase supply (415 V, 50 Hz), two contactors (K1 for low speed and K2 for high speed), interlocks to ensure safety, meters for measuring voltage, current, and power, a tachometer for speed measurement, and a brake/dynamometer for applying load. Proper wiring, grounding, and interlocking were checked before running the motor.

The basic formula for synchronous speed is:
$Ns = 120f / P$
where f is supply frequency and P is number of poles. When the motor runs, the rotor speed is always slightly less than synchronous speed. Slip is calculated as:
$s = (Ns - Nr) / Ns$

This tells us how much slower the rotor is compared to the synchronous speed.

To perform the experiment, the motor was first run in low-speed mode by switching on contactor K1. After the motor reached steady speed, voltage, current, power, and rotor speed were recorded at different loads such as 0%, 25%, 50%, and 75%. Then the motor was stopped completely. After that, contactor K2 was used to run the motor in high-speed mode, and the same readings were taken again under various loads.

The readings taken were filled into an observation table with voltage, current, input power, rotor speed, and torque for both speed modes. Using these values, synchronous speed, slip, torque, and efficiency were calculated. The results clearly showed two different speeds for the motor. High-speed mode consumed more power but also gave higher output. With increase in load, the current and slip also increased.

From the experiment, we understood that pole switching is an effective method to obtain two fixed speeds from the same motor without using a VFD. The Dahlander connection provides this advantage by changing the pole configuration. The experiment also showed the importance of interlocking, which ensures that both contactors cannot turn on at the same time, preventing damage.

Some important safety precautions include: do not energize both contactors together, always stop the motor before switching speeds, ensure proper grounding, keep hands away from rotating parts, and check all connections properly before starting.

Working Principle of Dahlander Motor

A Dahlander motor works on the principle of changing the number of poles in the stator winding. Normally, an induction motor runs at a speed close to its synchronous speed, which depends on the number of poles. In Dahlander motors, the stator winding is specially designed so that the same coil groups can be connected differently to create two pole numbers. This allows the motor to run at two fixed speeds, usually in a ratio of 2:1. Because the motor does not need an external speed controller, the Dahlander connection is simple, efficient, and reliable. It is often used in applications where only two constant speeds are required.

When the motor is connected in the low-speed configuration, more poles are formed, so the magnetic flux rotates slower, and the rotor also rotates at a lower speed. In the high-speed configuration, fewer poles are produced, so the flux rotates faster, causing the rotor to run at a higher speed. Although the winding remains the same, only its connection changes, making the Dahlander system economical and compact.

Applications of Pole Switching

Dahlander motors are widely used in industries where machinery requires two fixed speeds for different stages of operation. Common applications include:

- Industrial fans, where low speed is used for normal ventilation and high speed for heavy air movement.

- Pumps, especially centrifugal pumps, where speed affects the discharge rate.
- Wood-cutting machines, where different materials require different speeds.
- Machine tools that require rough machining at low speed and finishing at high speed.
- Compressors where varying air delivery is needed. These applications benefit from the simple and robust two-speed control system provided by the Dahlander motor.

Advantages of Dahlander Motor

One major advantage is that the Dahlander motor provides two speeds without the need for electronic devices such as a variable frequency drive (VFD). This makes the system cheap, reliable, and easy to maintain. The control circuit is straightforward and does not require complex programming. Since the winding is the same for both speed levels, the motor remains compact, reducing cost and space requirements. Power dissipation is minimal, and switching between speeds is fast. Additionally, the motor offers good efficiency and torque characteristics depending on whether it is connected in constant-torque or constant-power mode.

Disadvantages and Limitations

However, the Dahlander motor also has some limitations. It provides only two fixed speeds, so it cannot offer smooth or continuous speed control like a VFD-based system. Switching must be done carefully, preferably when the motor is stopped, to avoid mechanical shock or electrical damage. The motor design is more complex than a regular induction motor, and rewinding or repairing the Dahlander windings requires special knowledge. It is not suitable for applications needing frequent or rapid speed changes. Also, the torque characteristics change significantly between low and high speeds, which may not be suitable for all loads.

Detailed Step-by-Step Operation

When the motor is started in low speed, the contactor K1 energizes and connects the windings in a way that produces a higher number of poles. The motor accelerates gradually to its lower operating speed. Once the readings are taken, the load is adjusted using the brake or dynamometer to study the performance. After completing the low-speed section, the motor is stopped. This is important because switching directly from low to high speed can harm the winding and contactors.

Once the motor is completely at rest, contactor K2 is energized. This connects the windings to form fewer poles, allowing the motor to run at a higher synchronous speed. The motor reaches its higher rated speed, after which readings are again recorded at different load levels. This helps in understanding performance characteristics like how current, slip, and power vary between the two speed settings.

Graphs and Their Importance

In a detailed report, graphs help in understanding the motor behavior more clearly. The main graphs that can be plotted from the experiment data are:

- Speed vs Load
- Torque vs Speed
- Current vs Load
- Efficiency vs Load These graphs show how the motor responds to changing conditions. For example, as the load increases, the current also increases, and the speed slightly decreases due to increasing slip. Comparing graphs of low speed and high speed gives a clearer picture of how pole switching affects motor performance.

Extended Conclusion

From the extended study of the Dahlander motor, it becomes clear that pole-switching technology offers an effective and economical way to achieve two-speed operation in industrial equipment. The experiment demonstrates that the motor operates reliably at both low and high speeds, with performance characteristics closely matching theoretical expectations. The synchronous speeds calculated using the formula align well with the observed rotor speeds. Slip increases with load, as expected for induction motors. The use of interlocks ensures safe transitions between speeds. Although the system lacks continuous speed control, its simplicity and robustness make it suitable for many industrial applications.

Overall, the Dahlander motor remains a practical solution wherever two-speed operation is required.